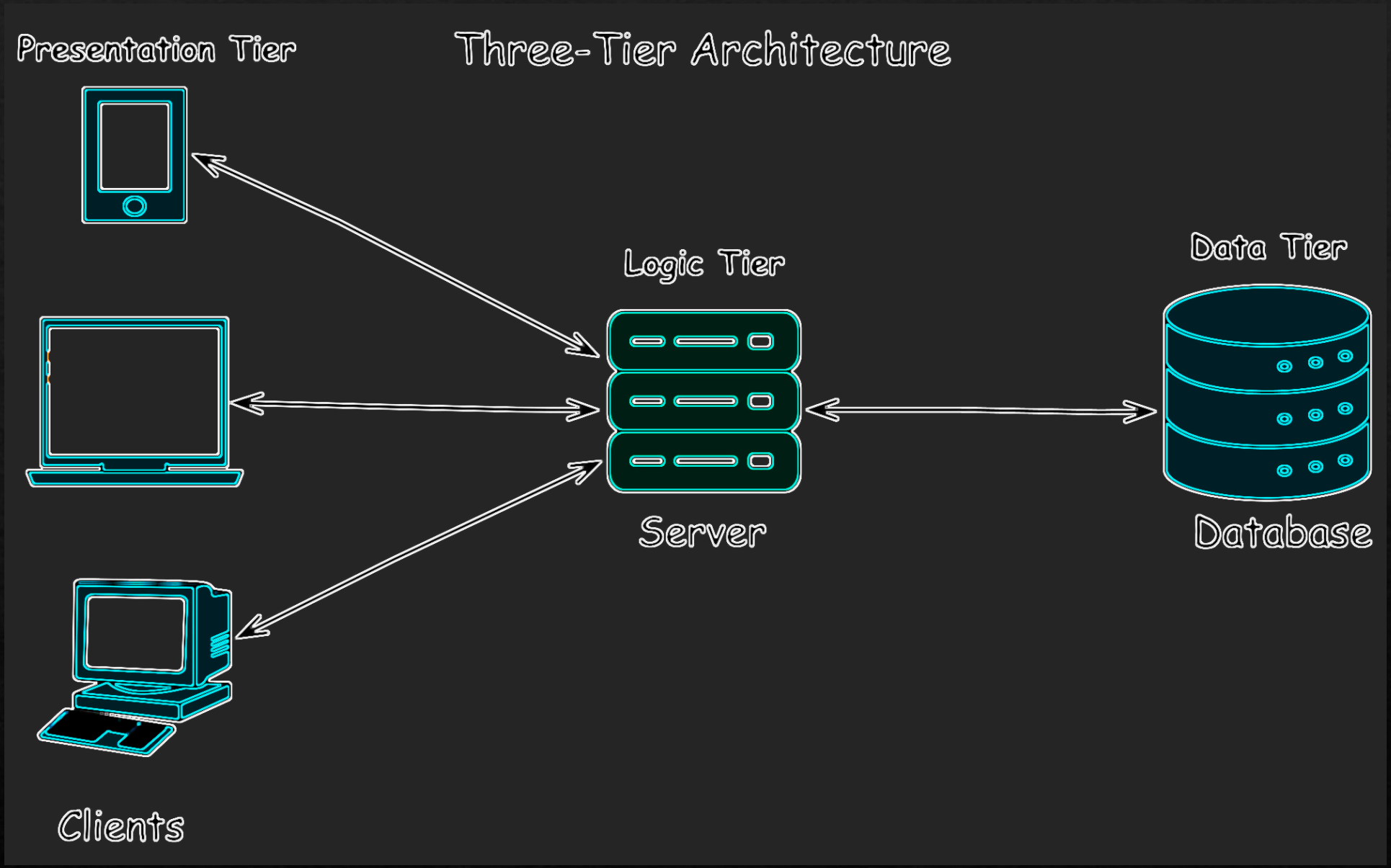
Basic Architecture Of Full Stack Project

What is a Full-Stack Project?

A full-stack application has 3 main layers:

- Frontend (Client)
- Backend (Server)
- Database

These 3 talk to each other in a fixed workflow.

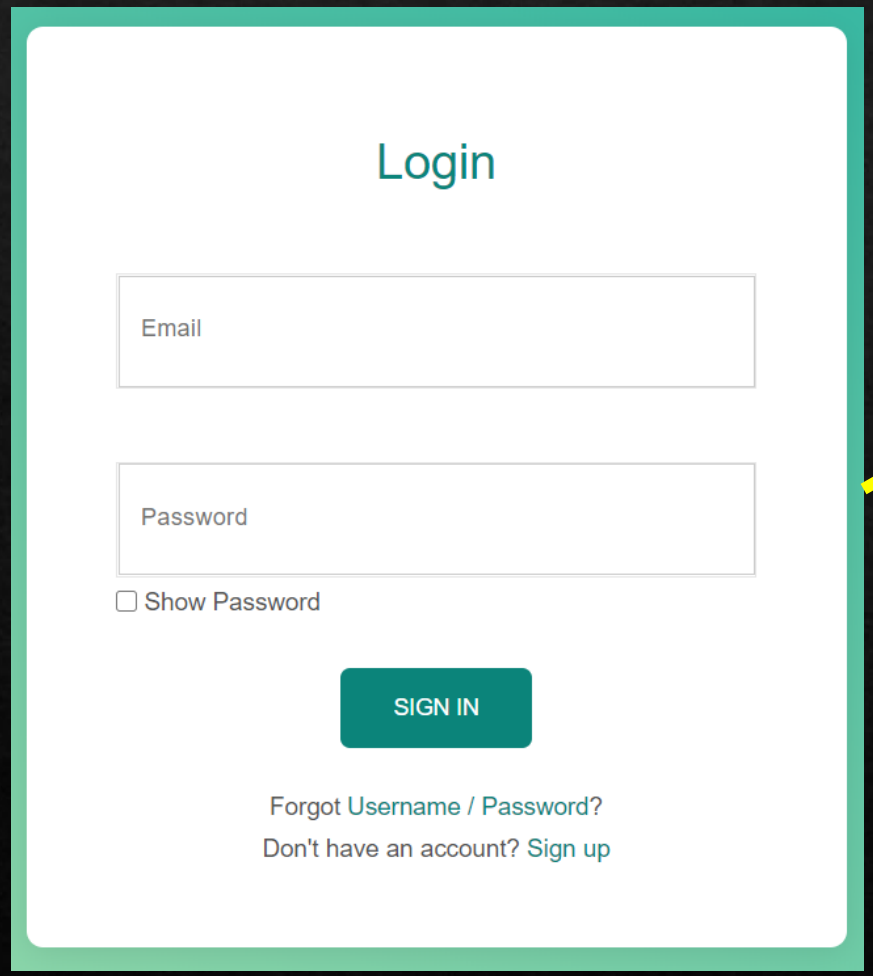

Presentation Tier
Three-Tier Architecture
Logic Tier
Data Tier
Server
Database
Clients

Frontend (Client Side)

- Runs in browser
- Built using React.js in MERN
- Responsible for UI + UX

What frontend does

- Shows UI (buttons, forms, pages)
- Takes user input (login, signup, post, comment)
- Sends data to backend via HTTP requests
- Displays response from backend



Backend Server
(Logic Tier)

Backend (Server Side)

- ❑ Runs on a server
- ❑ Built using Node.js + Express.js
- ❑ Acts as brain of the application

What backend does

- ❑ Receives request from frontend
- ❑ Validates data
- ❑ Applies business logic
- ❑ Talks to database
- ❑ Sends response back to frontend

Backend responsibilities

- ❑ Authentication (login / signup)
- ❑ Authorization (who can do what)
- ❑ Validation
- ❑ Security
- ❑ APIs
- ❑ Data processing

Database (Data Layer)

- Stores application data permanently
- In MERN → MongoDB

What database does

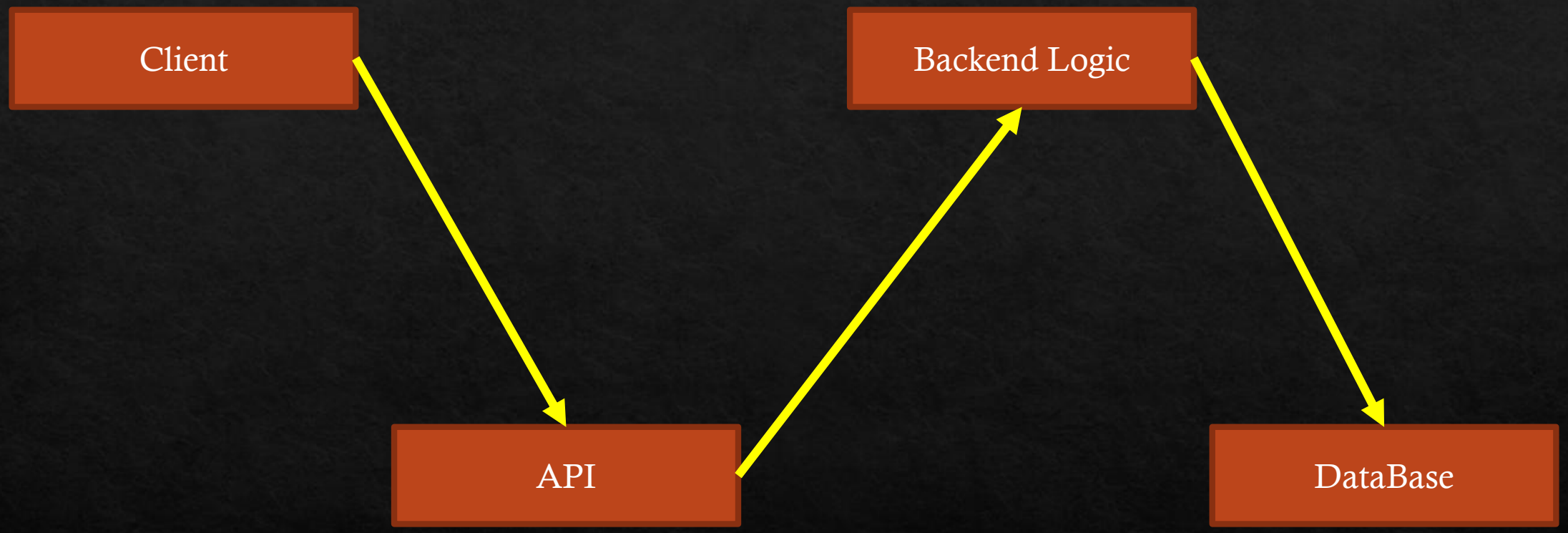
- Stores user data in a structured manner.

Database responsibilities

- Data Definition
- Data Storage
- Data Retrieval
- Data Manipulation
- Data Integrity
- Concurrency Control
- Security & Access Control:

Complete Workflow Example:

1. User opens website
2. React UI loads in browser
3. User performs an action (click, submit)
4. React sends HTTP request to Backend
5. Backend receives request
6. Backend processes logic
7. Backend talks to MongoDB
8. MongoDB returns data
9. Backend sends response
10. React updates UI



Just a URL + method (get, put, etc)

Dialogue of the video:

“Frontend kabhi directly database se baat nahi karta —
sirf backend ke through.”

Thank You